Creating a Basic ASP.NET Core MVC Web App

Getting Started & Structure

Step 1: Create the MVC Project

- Open **Visual Studio**
- Create a new project
- Select **ASP.NET Core Web App (Model–View–Controller)**
- Choose **.NET 5**
- Create the project

Step 2: Understand the MVC Structure

- **Controllers** → Handles requests
- **Models** → Holds data
- **Views** → Displays data
- **wwwroot** → CSS, JS, images

Models and Controllers

Step 3: Create a Model

`Models/Student.cs`

```
public class Student { public int Id { get; set; } public string Name { get; set; } public string Course { get; set; } }
```

Step 4: Create a Controller

`Controllers/StudentController.cs`

```
using Microsoft.AspNetCore.Mvc; public class StudentController : Controller { public IActionResult Index() { return View(); } }
```

Views and Data Flow

Step 5: Add a View

`Views/Student/Index.cshtml`

```
@{ ViewData["Title"] = "Student Page"; } <h2>Student Information</h2> <p>Welcome to the Student Page</p>
```

Step 6: Pass Data from Controller to View

Controller Code:

```
public IActionResult Index() { Student student = new Student() { Id = 1, Name = "Juan Dela Cruz", Course = "BSIT" }; return View(student); }
```

View Code:

```
@model Student <h2>Student Details</h2> <p>ID: @Model.Id</p> <p>Name: @Model.Name</p> <p>Course: @Model.Course</p>
```

Routing and Filters

Step 7: Routing

Access the page using:

```
/Student/Index
```

MVC automatically maps:

- `StudentController`
- `Index()` action

Step 8: Add a Filter to the Web App

Apply a filter to the controller:

```
[ServiceFilter(typeof(LogActionFilter))] public class StudentController : Controller { public IActionResult Index() { return View(); } }
```

This logs activity every time the page is loaded.

Execution

Step 9: Run the Application

- Press **F5**
- Open browser
- Go to `/Student/Index`
- View output and logs